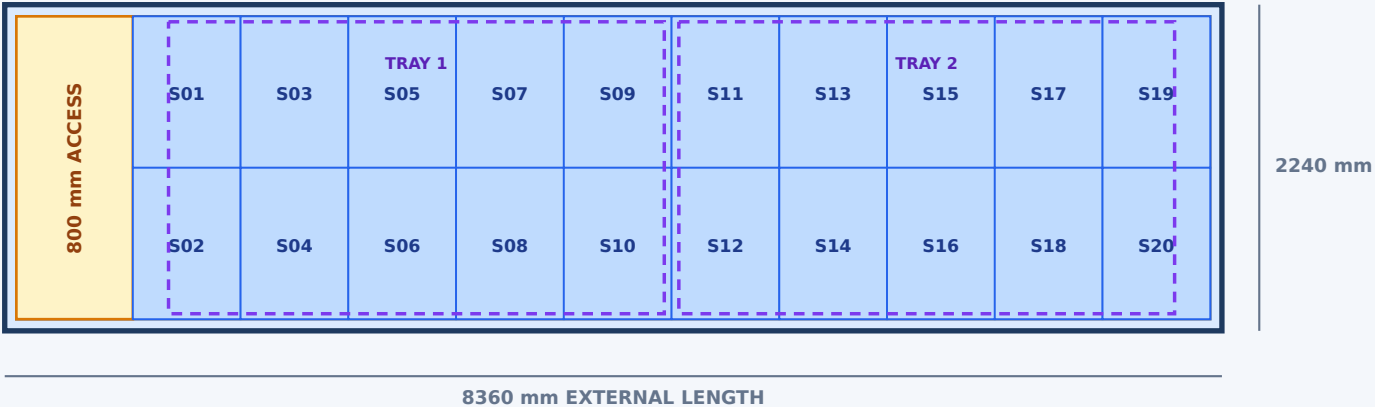


Bio/Degas Plenum

Arrangement & specification reference

Source-accurate coordination basis for a future 3D implementation

CURRENT IMPLEMENTATION + FUTURE 3D CONTRACT



58-CRATE WORKED EXAMPLE

8360 × 2240 × 2390 mm external envelope · 2 rows · 3-high nominal · 2 dispersion trays · capacity 60

Document status COORDINATION REFERENCE

Revision 01 · 22 September 2026 · Prepared for 3D project handoff

NOT A FABRICATION RELEASE

Verify structural, FRP, hydraulic, lifting, transport and code requirements before manufacture.

01 / READ FIRST

Purpose, scope & status key

This document translates the implemented Bio/Degas plenum designer into a self-contained arrangement and visualization contract. It is intended for engineering coordination and a separate 3D project—not for fabrication release.

IN SCOPE

Media-driven crate quantity; recommended arrangement; tank/tray/access dimensions; top and longitudinal section semantics; validation; persistence and stale-result behavior; future 3D scene mapping.

OUT OF SCOPE

Structural calculations, FRP laminate schedule, hydrostatics, hydraulics, drainage/overflow, lifting, transport certification, fire/code approval and fabrication tolerances.

Status legend

Badge	Meaning	Use in this reference
IMPLEMENTED	Present in current React/SVG designer or server contract.	Treat as current behavior.
DERIVED	Equation or value calculated from current constants.	Reproduce exactly.
FUTURE 3D	Recommended scene/interaction translation.	Do not claim it exists today.
CONFIRM	Engineering decision outside this tool.	Requires project review.

READING RULE

Solid linework represents current implemented visual semantics. Dashed or “FUTURE 3D” badges identify a proposed translation only. The source of truth remains the current media result, saved layout and server validation.

Source: BioDegasPlenumDesigner.tsx; bio-degas-plenum-crate-sizing.md; bio-degas-plenum-product-boundary.md

02 / INPUTS

Design basis & governed constants

The current media result is the only sizing input. The UI resolves the first finite value in this precedence order:

Precedence	Field	Conversion
1	preview.media.installed_media_m3	Use m³ directly
2	preview.media.installed_media_l	Divide litres by 1000
3	preview.media.required_media_m3	Use m³ directly
4	preview.media.required_media_l	Divide litres by 1000

Governing constants

Parameter	Value	Role
Media capacity	0.238 m³ / crate	crate count denominator
Crate envelope	1040 L × 740 W × 530 H mm	L × W × H
Maximum stack	4 crates	hard vertical cap
Recommended stack	3 crates high	default candidate
Minimum wall	80 mm	external-to-internal allowance
Dispersion tray	3405 × 2005 × 620 mm	L × W × H
Tray separation	100 mm	between trays, longitudinal
Tray-to-crate gap	100 mm minimum	vertical clearance
End access	800 mm minimum	usable length deduction
Transport envelope	9000 × 2250 × 2500 mm	external L × W × H

KEY CONVENTION

The implemented assessment uses $\text{internalHeight} = \text{externalHeight} - \text{wall}$ (one wall subtraction), while internalLength and internalWidth subtract two wall thicknesses. The server is authoritative for finite, positive dimensions and all fit checks.

03 / CALCULATION

Crate sizing & recommendation algorithm

CRATE COUNT

$N = \text{ceil}(\text{mediaVolumeM3} \div 0.238)$. If the resolved value is non-finite or ≤ 0 , no designer is rendered.

Recommended candidate generation

- Fix wall = 80 mm and nominal stack = 3.
- $\text{positions} = \text{ceil}(N \div 3)$. For rows $r \in \{1, 2\}$, $\text{columns} = \text{ceil}(\text{positions} \div r)$.
- $\text{candidateLength} = \max(3405 + 800 + 2 \times 80, \text{columns} \times 740 + 800 + 2 \times 80)$.
- $\text{candidateWidth} = \max(2005 + 2 \times 80, \text{rows} \times 1040 + 2 \times 80)$.
- Discard candidates above 9000×2250 ; sort by area, then length, then rows. Select the first.
- If no candidate fits, fallback is rows=2, length=9000, width=2250.

Derived recommendation fields

Field	Formula / rule
tray_count	2 iff $\text{internalWidth} \geq 2005$ and $\text{internalLength} - 800 \geq 2 \times 3405 + 100$; otherwise 1
external_height	$\min(2500, 80 + 3 \times 530 + 100 + 620) = 2390 \text{ mm}$
water_level	$3 \times 530 = 1590 \text{ mm}$ for recommended layout
stack positions	$\text{row} = \text{index} \bmod \text{rows}$; $\text{column} = \text{floor}(\text{index} \div \text{rows})$; $\text{quantity} = \min(3, N - \text{index} \times 3)$

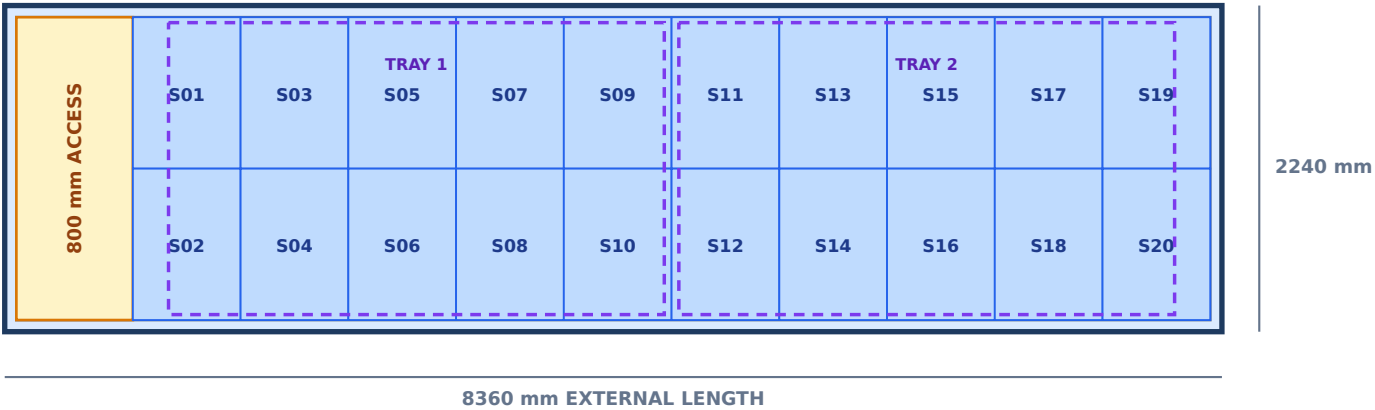
Source: `createRecommendedPlenumLayout` and `createStackPositions`

04 / REFERENCE CASE

Worked example — 58 crates

This is the governed regression example used by the current tests and is the preferred visual reference for the first 3D scene.

Check	Calculation	Result
Media basis	$\text{ceil}(13.804 \div 0.238)$	58 crates
Nominal arrangement	2 rows × 10 positions × 3 high	20 positions / nominal 60
Actual stack quantities	$18 \times 3 + 2 \times 2$	58 crates
Recommended external	candidate selection	8360 × 2240 × 2390 mm
Internal basis	$L=8360-160$; $W=2240-160$; $H=2390-80$	8200 × 2080 × 2310 mm
Vertical fit	$2310-620-100 = 1590$; $\text{floor}(1590 \div 530)$	3 layers
Capacity	crate positions × min(3,3)	60 crates
Trays	$2 \times 3405 + 100$	6910 mm occupied
Access	end allowance	800 mm
Validation	wall / envelope / tray / quantities	PASS / no errors



X / longitudinal

Y / transverse

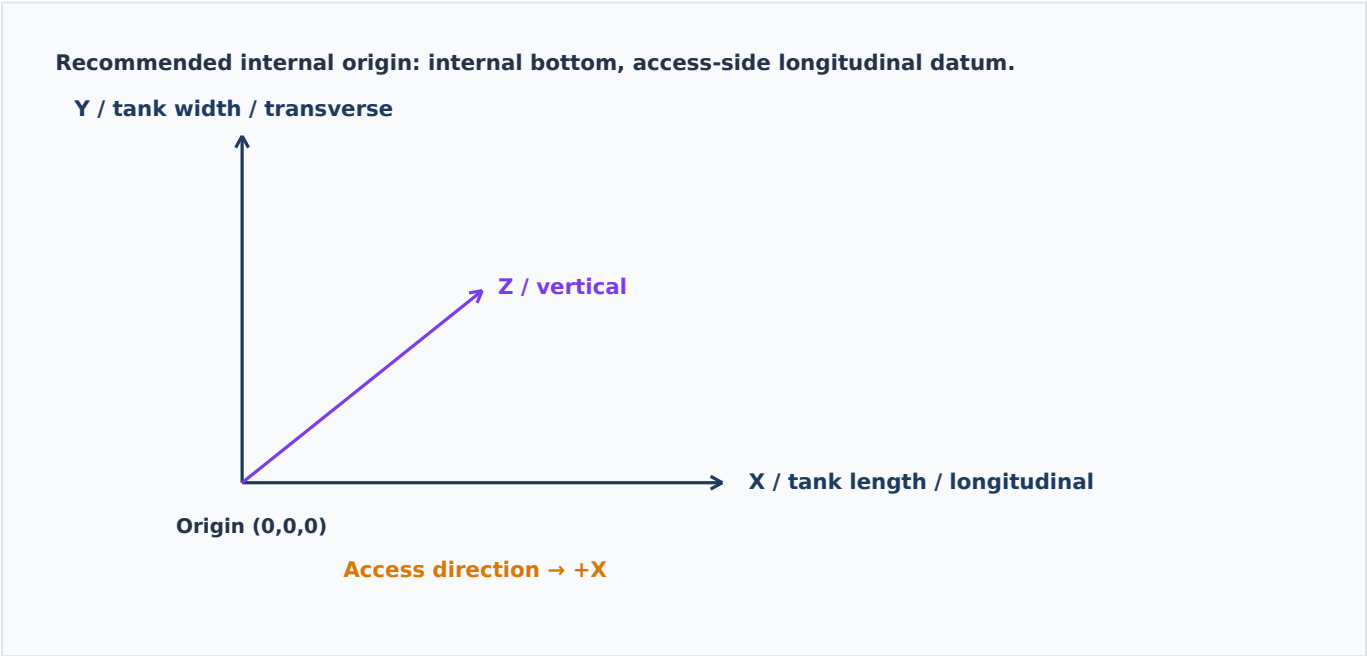
PASS BASIS

Wall 80 mm ✓ · tray gap 100 mm ✓ · two trays fit ✓ · capacity $60 \geq$ required 58 ✓ · stack quantities reconcile ✓

05 / SPATIAL CONTRACT

Coordinate system & datums

Future 3D must preserve the semantic axes already implied by the SVG. The coordinate origin below is a recommended internal datum; confirm external datum and fabrication datums with the project team.



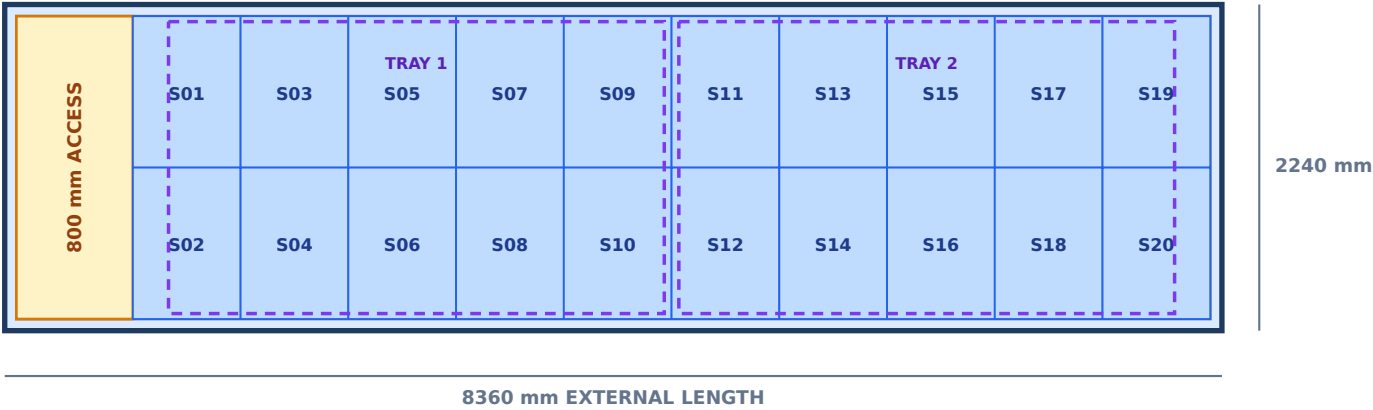
Coordinate / entity	Placement rule
Tank internal box	$0 \leq X \leq \text{internalLength}$; $0 \leq Y \leq \text{internalWidth}$; $0 \leq Z \leq \text{internalHeight}$
Stack S(i)	$X = \text{wall} + \text{actualAccessLength} + \text{column} \times 740$; $Y = \text{wall} + \text{row} \times 1040$
Crate level	$Z \text{ base} = \text{level} \times 530$; level 0 sits on internal bottom; lower crates remain fixed
Tray start	$X = \text{wall} + \text{actualAccessLength} + \max(0, (\text{occupiedEquipmentLength} - \text{totalTrayLength}) / 2)$
Tray vertical	tray bottom $z = \text{internalHeight} - \text{trayHeight}$; tray top $z = \text{internalHeight}$
Water	Internal footprint; top at internal bottom + <code>water_level_mm</code>

06 / IMPLEMENTED VISUAL

Top view — plan arrangement

IMPLEMENTED

The current top SVG shows external shell, internal clear envelope, amber access zone, stack rectangles and purple tray outlines. It is not a 3D model.



X / longitudinal

Y / transverse

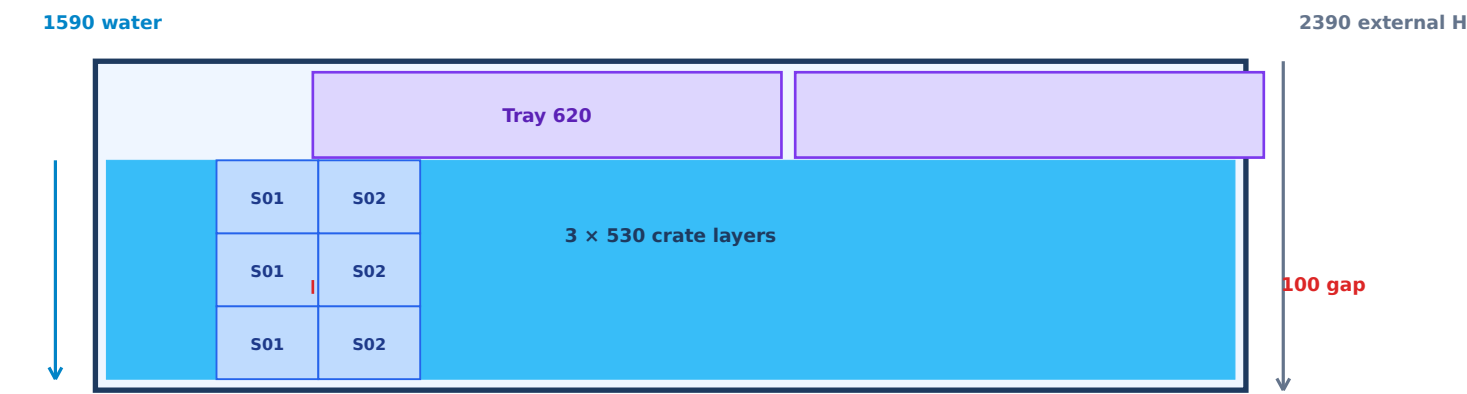
Visual element	Current rendering semantic	3D handoff
Shell	Blue #dbeafe, navy #1e3a5f outline	Rectangular FRP shell placeholder
Access	Amber #fef3c7 / dashed #d97706	Non-equipment service clearance
Stacks	740 × 1040 plan rectangles; x by column, y by row	Instance crate meshes by stack metadata
Trays	3405 × 2005 purple dashed outlines; 100 separation	Solid tray volumes
Dimensions	External length/width callouts	Dimension annotations / measurement tool

07 / SECTION

Longitudinal section & vertical stack

IMPLEMENTED

Section view filters to one selected row and sorts stacks by column. It exposes tray, gap, water and crate-layer relationships.



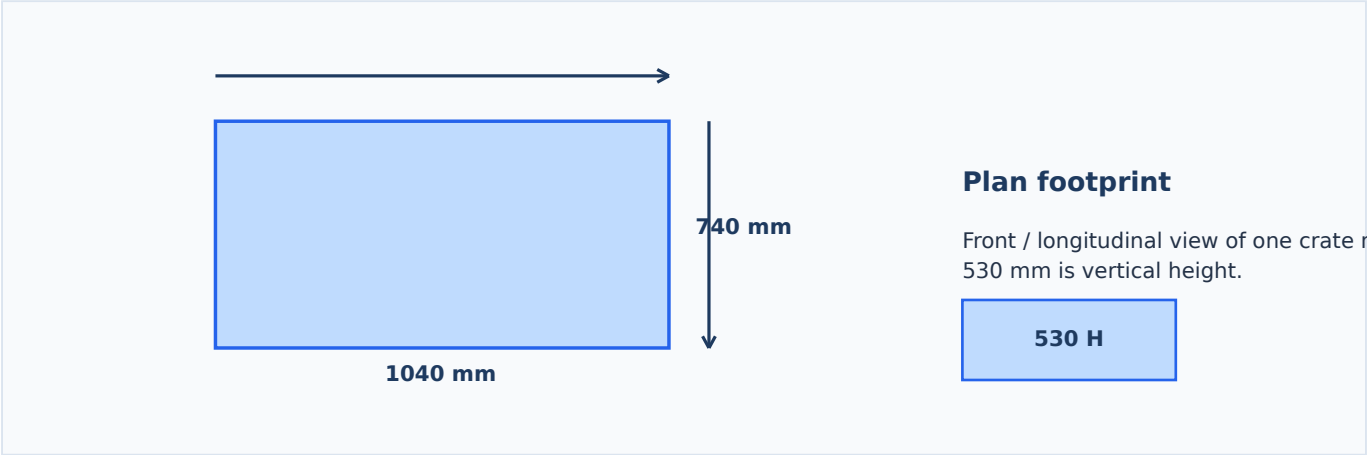
Implemented convention: $\text{internalHeight} = \text{externalHeight} - \text{wall} = 2310 \text{ mm}$

Vertical relation	Reference value	Interpretation
External height	2390 mm	recommended case
Implemented internalHeight	$2390 - 80 = 2310 \text{ mm}$	one-wall subtraction
Tray height	620 mm	tray bottom $z=1690$; top $z=2310$
Crate levels	$z=0, 530, 1060 \text{ mm}$	bottom-anchored from internal $z=0$
Crate top	$3 \times 530 = 1590 \text{ mm}$	top of level 2
Tray-to-crate gap	$1690 - 1590 = 100 \text{ mm}$	clearance below tray
Water level	1590 mm	bounded $0 \dots \text{internalHeight}$; visual fill only

08 / COMPONENTS

Crate orientation, indexing & placement

The crate envelope is length × width × height = 1040 × 740 × 530 mm. In plan, the 1040 mm dimension runs across rows (Y); the 740 mm dimension advances columns (X).



Index	row	column	quantity
S01	0	0	3
S02	1	0	3
S03	0	1	3
S04	1	1	3
...	index mod 2	floor(index ÷ 2)	min(3, remaining)
S19	0	9	2
S20	1	9	2

INTERACTION

Clicking or keyboard-selecting a stack highlights it and sets the section row. Changing rows or nominal stack height regenerates stack positions and can discard manual positions/quantities.

09 / PLACEMENT

Tray, access & clearance equations

Plan placement

Symbol	Implemented equation
internalLength	$\text{externalLength} - 2 \times \text{wall}$
internalWidth	$\text{externalWidth} - 2 \times \text{wall}$
occupiedCrateLength	$\text{max}(\text{column} + 1) \times 740$
totalTrayLength	$\text{trayCount} \times 3405 + \text{max}(0, \text{trayCount} - 1) \times 100$
occupiedEquipmentLength	$\text{max}(\text{occupiedCrateLength}, \text{totalTrayLength})$
actualAccessLength	$\text{max}(0, \text{internalLength} - \text{occupiedEquipmentLength})$
trayStartX	$\text{wall} + \text{actualAccessLength} + \text{max}(0, (\text{occupiedEquipmentLength} - \text{totalTrayLength}) / 2)$

Fit and vertical equations

Check	Equation / rule
crateLengthCapacity	$\text{max}(0, \text{floor}((\text{internalLength} - 800) \div 740))$
crateWidthCapacity	$\text{max}(0, \text{floor}(\text{internalWidth} \div 1040))$
verticalAvailable	$\text{internalHeight} - 620 - \text{trayGap}$
verticalCapacity	$\text{max}(0, \text{min}(4, \text{floor}(\text{verticalAvailable} \div 530)))$
capacity	$\text{crateLengthCapacity} \times \text{effectiveRows} \times \text{min}(\text{stack_height}, \text{verticalCapacity})$
tray condition	$\text{internalWidth} \geq 2005 \text{ and } \text{internalLength} - 800 \geq \text{totalTrayLength}$

IMPORTANT

The nominal 800 mm access is a minimum deduction in capacity and tray fit. The visual actualAccessLength is recomputed from the occupied equipment footprint and may be larger.

Validation, state & persistence

Validation matrix

Layer	Checks
Client assessment	wall ≥80; external limits; tray count 1/2; tray fit; gap ≥100; stack height ≤ vertical capacity; capacity ≥ required; grid bounds; no duplicate row:column; quantities 1...limit; quantity sum; water bounds
Server authority	finite positive dimensions; wall*2 < length/width; rows 1/2; stack 1...4; trays 1/2; gap ≥100; tray fit; capacity; nonempty valid stacks; source crate count recomputed from current media
Human confirmation	structural loads; FRP laminate; hydrostatic pressure; hydraulics; drainage/overflow; access ergonomics; lifting; transport; code compliance

State and persistence

- React Query key: bio-degas-plenum-layout + systemId + bookId; GET is enabled only with both IDs and a positive crate count.
- Saved layout is merged over a fresh recommendation; missing stacks are regenerated. Recommended resets local layout.
- Save is blocked by assessment errors or stale media. PUT includes design_book_id and source_crate_count, but server recomputes the source.
- Saved layout creates/updates one project equipment item and one generated plenum line plus a separate tray line quantified by tray_count.
- Draft/In Progress reads live data; For Review/Approved/Released reads the corresponding snapshot. Database triggers prevent mutation of frozen design-book states.

STALE RESULT

Staleness = preview.is_stale OR saved.source_crate_count ≠ current crateCount. The UI preserves the saved layout but requires Biological Media recalculation before saving.

Future 3D contract & appendix

FUTURE 3D RECOMMENDATION

No 3D scene, camera, mesh or WebGL implementation exists in the current application. The hierarchy below is a translation contract for a separate project.

Scene node	Responsibilities
Scene / PlenumTank	External L×W×H; shell placeholder; wall thickness metadata
InternalVolume	Clipping volume and internal coordinate frame
AccessZone	Actual access length; amber conceptual clearance
EquipmentZone	Container for trays and crate stacks
DispersionTrays / Tray_1...N	3405×2005×620 solids; longitudinal 100 gap
CrateStacks / Stack_S01...	Instances from row, column, quantity; 740×1040×530 each
WaterVolume	Internal footprint; level above internal bottom
Dimensions / Annotations	L/W/H, wall, access, gap, trays, water and stack labels
Selection / InteractionOverlay	Stack highlight, row section filter, clearance diagnostics

PLACEMENT CONTRACT

$x = wall + access + column \times 740$; $y = wall + row \times 1040$; $z_{base} = level \times 530$ from internal bottom. Tray bottom $z = internalHeight - trayHeight$; tray top $z = internalHeight$. Preserve lower crate elevations when quantity decreases. Future 3D features remain recommendations.

Quick reference / glossary

Term	Definition
External envelope	Tank outside dimensions; transport limit applies
Internal envelope	Length/width subtract two walls; height subtracts one wall in implemented assessment
Stack position	Unique row:column top-view coordinate
Tray gap	Vertical tray-to-crate minimum, 100 mm
Access length	End clearance deduction, 800 mm minimum
Water level	Bounded visual fill measured from internal bottom
PIM promotion	Explicit creation of reusable FRP tank product from immutable source snapshot